

Microfluidics for bacterial imaging

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1 INTRODUCTION

The past two decades have seen great advances in microscopy technology, especially the development of microscopes with high and ‘super’ (subdiffraction limit) resolutions. Complementing the advent of high-power microscopes is the availability of new or improved methods for labelling cellular targets. In the field of microbial research, there is an ever-increasing need for live cell imaging and long-term examination of living cells in order to observe cellular processes and biological machinery in action, and to track changes at the single-cell level as well as at a multicellular/population scale.

One of the commonly used methods for live bacterial cell imaging is to mount bacteria on agarose pads, which helps to immobilise the cells. Agarose pads infused with culture media will also support bacterial growth for a certain period of time. However, over time nutrients and oxygen will become limited, or the agarose pads dry out. Furthermore, many studies require growth media to be changed, or bacteria to be free swimming. The application of microfluidics to bacterial imaging not only resolves some of these issues but also opens new avenues for research, by creating and maintaining microenvironments to meet the specific needs of experiments or bacteria.

Microfluidics is the science and technology of manipulating fluid flow in geometrically constrained spaces, usually networks of channels and shaped chambers, at a microscale (Whitesides, 2006; Whitesides, Ostuni, Takayama, Jiang, & Ingber, 2001). The field emerged in the 1980s, when microelectromechanical systems (MEMS), tiny integrated devices or systems that combine mechanical and electrical components, were first manufactured. Subsequently, these were used in industries such as in pressure sensor and inkjet printer head manufacturing. In the 1990s MEMS started to be applied to the fields of biology, biomedicine and chemistry, initially for

analysis and then also for detection and other applications such as drug delivery, further fuelling the development of microfluidics. The best known are the Lab-on-a-Chip systems, sometimes also called Bio-MEMS or miniaturised total analysis systems (μ TASs), which are miniaturised devices that integrate one or several laboratory procedures on a single chip of submillimetre scale (Volpatti & Yetisen, 2014).

The field of microfluidics involves several disciplines including engineering, physics, chemistry, computer science, nanotechnology, biology and medicine. Microfluidics can reduce sample and reagent consumption and increase automation and parallelisation, helping to reduce cost as well as analysis time (Mark, Haeberle, Roth, von Stetten, & Zengerle, 2010). The past decade has seen a great increase and expansion in microfluidic applications, spanning from manufacturing, diagnostics, analysis, to research. In biology, microfluidic technology is widely used in genomic and proteomic analyses, as well as flow cytometry, detection and water analysis (Beebe, Mensing, & Walker, 2002; Bennett & Hasty, 2009; Liu & Singh, 2013; Streets & Huang, 2014; Velve-Casquillas, Le Berre, Piel, & Tran, 2010; Whitesides, 2006; Whitesides et al., 2001).

One field that has benefited immensely from the development of microfluidics is the microscopic study of microorganisms, which is being revolutionised with very positive impacts (Hol & Dekker, 2014; Saleh-Lakha & Trevors, 2010; Weibel, Siegel, Lee, George, & Whitesides, 2007; Wessel, Hmelo, Parsek, & Whiteley, 2013). The technology allows bacteria to be grown in chemically and physically controlled environments, monitored in situ, in a noninvasive manner, while generating automated, real-time data on the behaviour and morphology of the cells with single-cell resolution (Balagaddé, You, Hansen, Arnold, & Quake, 2005).

One of the earliest bacterial studies using microfluidic microscopy was reported in 2004, where Balaban, Merrin, Chait, Kowalik, and Leibler (2004) created a microfluidic device with channels in which cells were forced to grow, thereby constraining the movement of the daughter cells and allowing the lineage of each cell to be traced. Furthermore, fluid flow in the device provided the cells with media and antibiotics in a controlled manner. Using this technique the authors were able to identify persister cells and show that persistence was linked to preexisting heterogeneity in bacterial populations. Examples of further applications of microfluidic microscopy in bacterial research are listed in Table 1.

While microfluidics has presented new opportunities for bacterial imaging, the broadening application of microfluidics in biological research is also driving rapid development of the technology. Several commercial microfluidics systems for microscopy are now available and have relatively standard protocols defined by the manufacturers (e.g. the BioFlux system and the CellASIC System). However, many applications will require specially designed microfluidic setups tailored for specific applications. In this chapter, we outline the general procedures involved in fabricating, setting up and controlling microfluidic devices (Sections 2–5) and review some recent microfluidic applications in bacteria imaging, focusing on research in the engineering of bacterial systems (Section 6), multispecies biofilm research

Table 1 Examples of Microfluidic Microscopy in Bacterial Research

Application	References
Persistence and persisters	Balaban et al. (2004), Wakamoto et al. (2013), Maisonneuve and Gerdes (2014)
Chemotaxis	Ahmed, Shimizu, and Stocker (2010), Rusconi, Guasto, and Stocker (2014), Chen et al. (2015)
Bacteria detection	Gómez-Sjöberg, Morissette, and Bashir (2005), Boedicker, Li, Kline, and Ismagilov (2008)
Single-cell gene expression	Longo and Hasty (2006), Taniguchi et al. (2010)
Gene expression regulation	Wallden and Elf (2011)
Chromosome behaviour	Larson et al. (2006), Pelletier et al. (2012)
Bacterial interactions	Kim, Boedicker, Choi, and Ismagilov (2008), Wessel et al. (2013)
Motility	Kaehr and Shear (2009), Ducret, Théodoly, and Mignot (2013)
Control of bacterial development	Norman, Lord, Paulsson, and Losick (2013)
Bacterial physiology	Campos et al. (2014), Taheri-Araghi, Brown, Sauls, McIntosh, and Jun (2015)
Biofilm formation	Nance et al. (2013), Kolderman et al. (2015), Oliveira et al. (2015)
Quorum sensing	Boedicker, Vincent, and Ismagilov (2009), Jeong, Jin, Lee, Kim, and Lee (2015)
Stress adaptation	Wang et al. (2014), Mathis and Ackermann (2016)
Identifying new bacterial species	Williams et al. (2016)
Single-molecule based super-resolution microscopy (smSRM)	Cattoni, Fiche, Valeri, Mignot, and Nöllmann (2013)
Host–pathogen interactions	Kim, Hegde, and Jayaraman (2010), Pamp, Harrington, Quake, Relman, and Blainey (2012)

and microbial ecology (Section 7), bacterial cell cycle and size homeostasis (Section 8) and cell shape and geometry (Section 9). Many of these applications used custom-designed microfluidic devices. We also provide step-by-step methods for two of these noncommercial devices (Section 11), both relatively easy to follow and adopt, with the aims of supplying practical information as well as to provoke ideas that can lead to new or further improved microfluidic devices.

2 FABRICATION OF MICROFLUIDIC DEVICES

Though there are a range of commercially available microfluidics devices available, there are many advantages to setting up a bespoke system. Although it takes time, the benefits of being able to design, build, test, feedback and improve the microfluidics devices are significant. This ‘do-it-yourself’ approach means that such devices can be more easily tailored to address specific research questions. Of course, there are

important considerations that are required to integrate microfluidics with microscope systems, such as the working distance and focal distance from the objective, through the base of the device and into the channels containing the cells.

A microfluidic chip is part of a microfluidic device in which microscaled features (microchannels and chambers) have been moulded, and therefore the most important component of the device. To avoid confusion, in this chapter we refer to the assembly that we mount on the microscope stage, mainly consisting of the microfluidic chip, the glass slide(s)/dish and any other accessories that form and support the channels and the chambers, as a microfluidic ‘device’. And the ‘device’ connected with the fluid controller and the microscope as a microfluidic ‘setup’. The basic steps of a typical process for fabricating a microfluidics chip, using soft lithography to produce a mould and then replica moulding with polydimethylsiloxane (PDMS) for the chip, can be seen in Fig. 1. Replica-moulded PDMS microfluidics chips are the predominant type of device used for bacterial applications (McDonald et al., 2000). Some of the elements of this process require specialised equipment. However, it is possible to simplify the process by outsourcing some of the more complex process steps, such as lithography of the silicon wafer and the production of the mask.

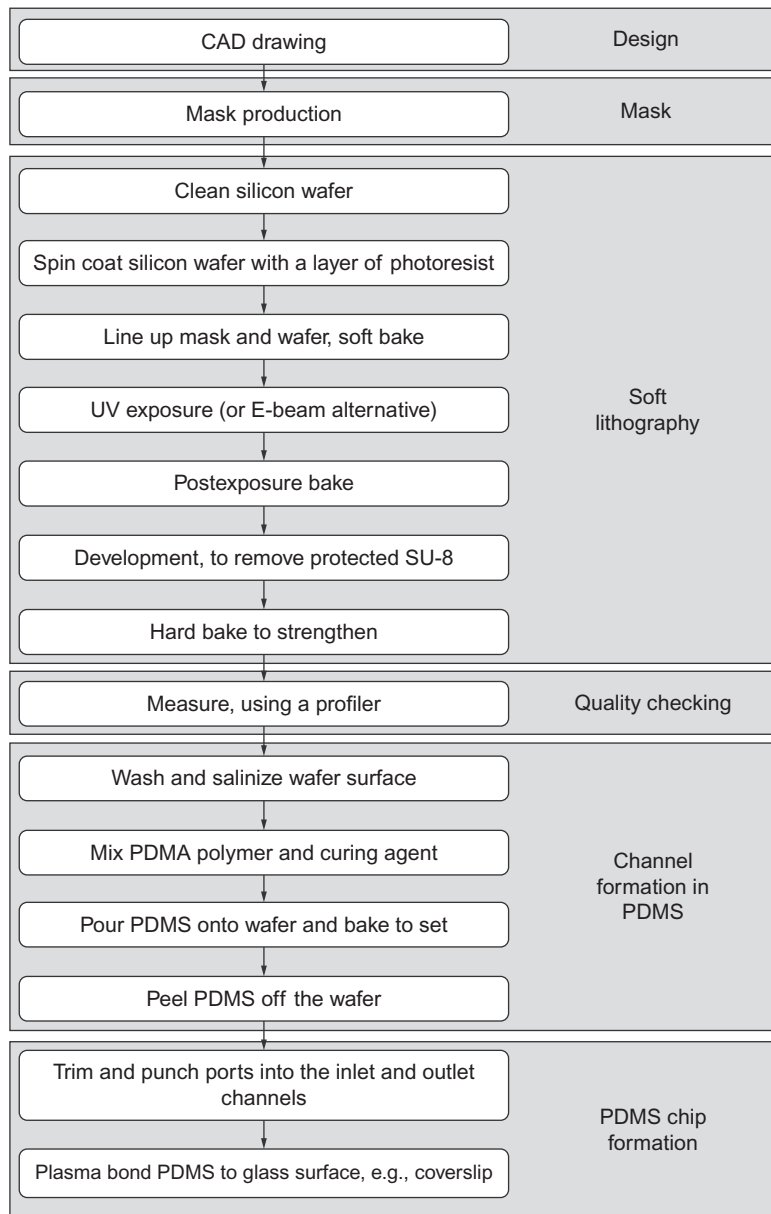
A number of compact and contained systems for the design and cost-effective production of microfluidics devices for rapid prototyping are also being developed (e.g. Boston University’s CIDAR lab). Methods vary but include the 3D printing of polymers (Dolomite Ltd., Royston, UK) or direct milling of hard plastic substrates using a CNC (computer numerical control milling machine) (e.g. MakerFluidics from CIDAR Lab, Boston University). However, most of these systems are limited to a minimum feature size of 5 µm or more depending on the system employed.

2.1 DESIGN

A number of design software options are available but typically Tanner L-edit software (Mentor Graphics, OR, USA) or AutoCAD software (AutoDesk, USA) is used to produce two-dimensional representations of the channel and chamber layout. The design will first need to be transferred onto a mask, for use later in the soft lithography stage. Masks are available from a number of companies, but require specific file formats, typically GDSII or CIF, and this needs to be checked before choosing the design software.

There are a number of useful guides available on the websites of some of the leading microfluidics laboratories on how to design the features in your microfluidics chip. The Stanford microfluidics foundry website (<http://web.stanford.edu/group/foundry/>) provides a wealth of useful design tips, as well as guides for using design software and some helpful design files.

As an alternative to traditional CAD software, a number of laboratories are developing design tools to assist with various parts of the design process, including Microfluidics design kit (PhoeniX software, The Netherlands) and Fluigi (Huang, 2015). The focus of these is to automate, speed up and simplify the process of designing functioning microfluidics.

**FIG. 1**

Flow chart of the main stages involved in producing a PDMS microfluidics chip, using soft lithography and replica moulding.

One of the key considerations when designing microfluidics channels and chambers is the aspect ratio, which is the ratio of height to width. Channels with an aspect ratio of less than 1:10 are likely to fail, with the channel collapsing in onto the bottom surface of the chip during bonding. Too large an aspect ratio ($>1:1$) can lead to failure in the lithography methods. These general rules vary according to the photoresist or production methods to be used, so it is important to check the specifications of the processes available at the beginning of the design process.

When the microfluidics designs involve two or more depths, regardless of when the two layers need to be aligned, two masks are needed. Both masks should include alignment marks, usually crosses incorporated at the design state, to facilitate their accurate alignment.

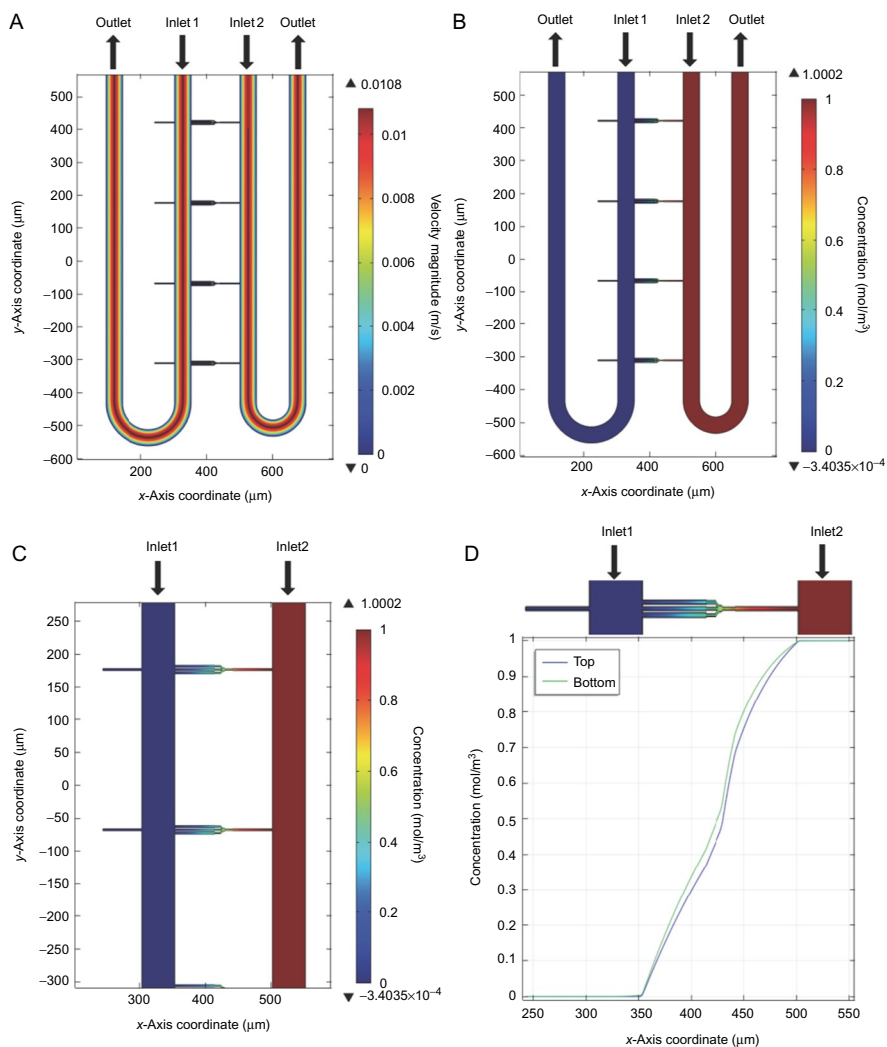
In order to design chips with the required fluid flow regime, the movement of liquid through microscale channels needs to be understood. Fluid dynamics at the scales relevant to microfluidics are markedly different to that at larger scales, with laminar flow and minimal mixing dominating. Laminar flow occurs when the Reynolds number is low (typically less than 1000), with the Reynolds number being defined as the inertial forces divided by the viscous forces acting within the channel. More detailed information on fluid flow in microfluidics can be seen in [Sharp, Adrian, Santiago, and Molho \(2005\)](#) and the Elveflow website (<http://www.elveflow.com/microfluidic-tutorials/>).

At this stage, before going into production, it is useful to model the flow regime and the behaviour of particles, nutrients, chemical concentrations and cells within the design. This ensures that the design is fit for purpose and allows the design to be modified if required preproduction. Modelling can be done with a number of software packages, such as COMSOL Multiphysics and FLOW-3D (FlowScience Inc., USA). COMSOL has a microfluidics package designed specifically to deal with flow through microscale structures. [Westerwalbesloh et al. \(2015\)](#) created a model using COMSOL, which enabled them to assess nutrient limitation in their microfluidics chip design. COMSOL has also been used by the authors of this chapter to model the fluid dynamics in a quorum sensing microfluidics assay chip, and the concentration of an inducer molecule applied to the chip through one of the inlets ([Fig. 2](#)).

2.2 PRODUCTION OF SILICON WAFERS

As seen in [Fig. 1](#), photolithography with ultraviolet light is frequently used to activate cross-linking in the photoresist compound beneath. However, the resolution and therefore the minimum feature size that can be achieved are limited by the wavelength of light used. When higher resolution is required for resolving smaller features, more expensive technologies, such as electron beams (E-beam) or ion beams, need to be used. This technology has been utilised for making microfluidics chips for the trapping and study of submicron-sized cells ([Moolman, Huang, Krishnan, Kerssemakers, & Dekker, 2013](#)).

An alternative to E-beam lithography is to use a standard photolithographic process and then use surface deposition to deposit a thin and even layer of a compound

**FIG. 2**

COMSOL simulation of fluid dynamics in a quorum sensing microfluidics assay chip, designed at Newcastle University by S. Park. (A) The liquid velocity profile of the fluidic channels given 70,000 Pa of pressure on the two inlets. (B) The concentration gradient of the fluidic channels when water is injected through inlet 1 and 1 mM concentration of inducer molecule is injected through inlet 2. (C) A close-up view of the concentration gradient field shown in (B). (D) Comparison of the concentration gradient across the *a*-axis between two locations (*Top* and *Bottom*) separated by about 6000 μm in the *y*-axis direction.

such as silicon oxide onto the surface, making the features smaller. This improves resolution so that channels can be made small enough to hold individual bacteria in a constrained position (Moffitt, Lee, & Cluzel, 2012) but without the cost of the E-beam methods.

The complexity of the channels and chambers required will determine the methods chosen for chip and silicon wafer fabrication. For example, if a chip has a number of features and all are the same depth, then a straightforward photolithography method using a layer of photoresist on the silicon surface is all that will be required. For more complex designs, where channels of different depths are required (e.g. where channels of a few microns depth lead off a deeper channel through which media flows), different methods need to be employed. There are two approaches to resolve this fabrication problem. The first option is to use photolithography to create the shallow channels in a photoresist layer and then etch again down into the silicon surface of the wafer itself for the deeper channels. The second option is to make two silicon wafers, using two separate masks: one mask for the shallow channels and the other for the deeper channels. The two layers are then replica moulded in PDMS at the required thickness, and the two layers aligned and bonded to produce the required chip. Alignment can be a challenge, as precision is required. This has led to the development of low cost in-house solutions to align wafers, such as that used by Li et al. (2015).

As mentioned earlier, the design needs to be translated onto the silicon wafer from which the PDMS microfluidic chip will be moulded (Fig. 1). This is done using an intermediate called a mask. Masks are typically made from quartz or soda-lime glass with a chromium surface or from celluloid film. Celluloid film is by far the cheapest mask material, but are less hard wearing and can only be used reliably for design features above 5 μm . On the other hand, glass masks are more expensive but can be used repeatedly with little degradation and for features down to approximately 0.8 μm . Both of these photomask types are suitable for use with UV photolithography.

Another important consideration for the transfer of a design onto a silicon wafer coated in a photoresist resin is whether the polymer used is a negative or positive photoresist, and how the chips will be moulded from it. A negative photoresist is a photoresist in which areas of the resist that are exposed to UV become cross-linked, and in all protected areas the photoresist can be washed away. When using a negative photoresist the mask is designed so that the area that will form the channel is transparent and the rest of the area is opaque (either by the printed or the chrome part of the mask). A commonly used negative photoresist is SU-8 (MicroChem Inc. and Gersteltec SARL), which can be used for the reliable production of features with a depth of between 1 μm and 1 mm. SU-8 is deposited onto the surface of the silicon wafer by centrifugation, with the rotational speed and time determining the thickness of the layer, which in turn determines the final depth of the microfluidic channel that is moulded from the wafer. New formations of SU-8 include SU-8 2000 (MicroChem Inc.; <http://www.microchem.com/pdf/SU-82000DataSheet2025thru2075Ver4.pdf>), which can be coated to make a minimum layer thickness of 0.5 μm , and SU-8

3000 (MicroChem Inc.; <http://microchem.com/pdf/SU-8%203000%20Data%20Sheet.pdf>) which can only be coated to 4 μm , but can be used to make structures with aspect ratios up to 5:1.

2.3 PRODUCTION OF THE MICROFLUIDICS CHIP

The next stage in the production of a microfluidics chip is to translate the microscale features from the silicon wafer onto the final chip material. Chips can be made from a range of materials including PDMS, SU-8 (Sikanen et al., 2007), agarose gel (Moffitt et al., 2012), glass and even paper (Tao, Xiao, Lei, & Lee, 2015). However, agarose and PDMS are most commonly used for bacterial applications, in conjunction with the soft lithography methods for patterning wafers. PDMS is an excellent material with which to produce microfluidic devices. It is inexpensive, optically transparent (down to approximately 280 nm) and has low autofluorescence (Piruska et al., 2005). PDMS is also nontoxic to microorganisms and is chemically inert (Mata, Fleischman, & Roy, 2005). PDMS can be used to reproduce submicron-sized features with high fidelity, so is ideal for its use in replica moulding from a patterned silicon-SU-8 wafer. PDMS is relatively easy to work with in most biological laboratories as it cures at low temperatures and is able to be covalently bonded to itself or other silica-based materials after plasma activation (McDonald et al., 2000). Due to its elastomeric properties, PDMS forms smooth surfaces, a feature that is very difficult to achieve when fabrication devices from harder and more brittle materials such as glass or acrylic. The flexibility and elastomeric properties of PDMS also make the formation of holes for inlet and outlet ports and measurement probes more straightforward. While media can flow through channels in PDMS, diffusion through the material itself is limited. In order to overcome this diffusion limitation, chips have been designed in which the central area used for growing and observing cells is made from patterned agarose gel. The patterned agarose pad can then be sealed into a PDMS support, and fluid flow can be directed through the delicate and soft agarose structures. One example of this type of chip is that designed by Moffitt et al. (2012). More details on the methods used to construct such agarose and PDMS devices are given in Section 11.

3 FLUID FLOW

Accurate, predictable and reliable control of fluid flow is vital for the smooth running of a microfluidics system. There are a number of decisions that need to be made including what type of pump to use to drive the flow and how to transmit the fluid from the pump into the chip.

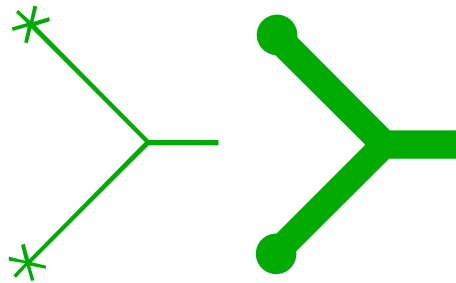
Three main types of pumps can be used to deliver media to microfluidics chips: peristaltic pumps, syringe pumps and pressure controllers. Peristaltic pumps are simple to use, but are less favoured for bacterial work, as they are not very accurate at low flow rates. Syringe pumps are relatively inexpensive and provide an easy to set

up solution to deliver media. Multiple syringe pumps connected to different inlet ports can provide not just media but also antibiotics or other chemicals when gradients are required. The amount of media delivered to a microfluidics chip using a syringe pump is easily measured and predictable. The main drawback of using this type of pump is that they deliver media in a pulsed manner and are not very responsive to changes in flow rate. Depending upon the resistance in the tubing, it can take up to an hour to equilibrate the flow rate to the required velocity. This leads to uncertainty in the amount of fluid being delivered during the time it takes for the flow rate to equilibrate, with external flow metres being used to estimate the actual volumes and rates of medium delivery. Additionally, if the microfluidic channel attached to the syringe pump becomes blocked, there is no feedback mechanism, so pressure will build up inside the device, usually leading to leakage from the ports or failure of the plasma bonding connecting the PDMS to the underlying glass surface. Either of these problems renders the chip unusable. An alternative to syringe pumps is to use a pressure-driven controller. Though these controllers tend to be more expensive, they are much more responsive to changes in flow rate and provide a much smoother, pulseless flow of fluid into the chip. Elveflow Plug and Play microfluidics provides a summary of the types of controllers available.

Transmission of the fluid from the pump into the microfluidic chip can be a problem as leaking can occur. The main causes of leakage are poorly fitting ports, blockage of the microfluidics channels (often with PDMS or cells) or too much pressure being applied from the pumps. In order to prevent leakage and blockage, a number of practical steps that can be taken. First of all, when ports are punched into PDMS, care must be taken to ensure that the core of PDMS is removed from the resulting hole; otherwise, this can be washed into and block the channels. Removing the core is made easier by using a biopsy punch that is slightly smaller than the size required for the post, for example, using a 0.75-mm diameter biopsy punch for a 0.8-mm port post. Thorough cleaning of the PDMS part of the chip and also the glass surface before plasma bonding will also help to remove any loose fragments of PDMS from the surface.

The other main cause of blocking is cell clumping. At the design stage, careful thought should be given to the design of the inlet and outlet channels and the buffer reservoirs. These channels can be made wider, and the chambers that form the reservoirs for the inlet and outlet ports can be made wider and rounder (Fig. 3, right), rather than narrow with cross-shaped ends, for example (Fig. 3, left). This reduces the number of small features in the vicinity of the ports for cells to get trapped in. It is also important to make sure that cell cultures or samples to be put into the channels are well mixed and in a single-cell suspension on entering the chip.

Valves can be a useful addition to the device if it is important to the experiment to have areas of the chip that can be separated off, or to control the precise timing or location of different components. This requires the use of a layer of pressurised air to control the flow at the valves. The control layer usually sits above the fluid layer, when using an inverted microscope. Valves can be designed in different ways,

**FIG. 3**

Left, an example of a port and inlet that would be more prone to clogging. The centre of the star is the position of the inlet hole and port. *Right*, a simplified version of the port with wider channels, which is less likely to clog. Designs drawn in L-edit software.

but those shown by [Grover, Ivester, Jensen, and Mathies \(2006\)](#) and [Unger, Chou, Thorsen, Scherer, and Quake \(2000\)](#) are the most commonly used for devices made with PDMS.

4 ADDITIONAL CONSIDERATIONS WHEN DESIGNING AND SETTING UP A MICROFLUIDICS SYSTEM

In addition to the control of fluids, nutrients and chemical gradients, gas exchange and temperature in the device are also important considerations. Gas exchange is largely driven by the type of material used to produce the device. PDMS is permeable to gas; however, this permeability can be affected by surface modifications, for example, plasma activation or attachment of proteins or chemical species. The thickness of the PDMS can also affect gas diffusion, with [Ochs, Kasuya, Pavesi, and Kamm \(2014\)](#) showing higher oxygen levels in 2 mm thick PDMS chips than in those that were 5 mm thick. It is possible to overcome any oxygen limitation in the medium by keeping it well aerated and increasing the rate at which the medium is replenished in the system. However, if a super oxygenated environment is required, such as when simulating a highly aerated environment, or if an anaerobic environment is required, then modifications would need to be made to the chip design. A number of solutions to this have been devised. [Lam, Kim, and Thorsen \(2009\)](#) designed a differential oxygenator, allowing the dissolved oxygen levels within the microfluidics device to be controlled. This system uses oxygen and nitrogen gas mixed in a microchannel in close proximity to the channels containing the cells. With this device the authors were able to culture organisms with different oxygen requirements. In order to maintain anaerobic conditions, [Steinhaus, Garcia, Shen, and Angenent \(2007\)](#) produced a PDMS microfluidics chip using the soft lithography method. The device was then encased in a small acrylic chamber sealed with an O-ring and silicon grease. All tubing leading to the device was also encased in neoprene tubing.

The simplest method for maintaining a set temperature is to have an external incubation chamber that covers both the microfluidic device and most of the microscope. The volume of air to be warmed in such incubation chambers is, however, relatively large, so it does not respond quickly to changes in the required temperature. External incubation chambers also tend to be an expensive solution if not already available in the lab. Preheating the input liquids is also an option. Commercial systems tend to have built-in systems that facilitate the preheating of the media and also have some form of incubation chamber (e.g. CellASICs, BioFlux). The review by [Miralles, Huerre, Malloggi, and Jullien \(2013\)](#) on heating and temperature control in microfluidic systems provides an excellent overview of the relative strengths and weaknesses of the various heating methods.

5 COMPUTATIONAL ANALYSIS AND CONTROL OF BACTERIAL MICROFLUIDIC SYSTEMS

Computing hardware and software are an integral part of a microfluidics system. They are used in the design of the systems themselves, their spatiotemporal control and the analysis of the bacteria. In many cases, there is a requirement for the operating systems to be controlled in real-time.

The small size of bacteria in comparison to eukaryotic cells presents challenges, not only in relation to the scale of microfluidics fabrication but also for the computational analysis, particularly for image analysis. Examples of some useful programmes for image analysis are shown in [Table 2](#). Tools such as MicrobeTracker ([Sliusarenko et al., 2011](#)) have proved valuable in this respect. MicrobeTracker allows outlines and segmented cells to be defined in images and minicolonies and also facilitates the tracking of cell lineage. An associated tool, SpotFinder, allows the foci of fluorescently labelled molecules inside bacterial cells to be tracked. Recently, MicrobeTracker and SpotFinder have been upgraded to Oufiti ([Paintdakhi et al., 2016](#)). However, in many cases the image analysis techniques that have been developed for bacterial microscopy are not optimal for microfluidics-based analysis, and novel approaches have been developed that are tailored to these devices.

For example, super-resolution fluorescence microscopy methods can be used in combination with microfluidics systems to investigate the architecture, composition and dynamics of bacterial substructures. [Cattoni et al. \(2013\)](#) describe an automated cell detection and image analysis system for single-molecule localisation in cell populations, facilitating the study of heterogeneity dynamics and other properties at a super-resolution level. This system also incorporates features that allow cell-flatness, stability and growth in microfluidics chambers to be monitored as well as the detection and characterisation of cell clusters.

While microfluidic systems offer a powerful approach to the study of bacterial motility, suitable computer-based analytical systems are required to interpret the complex images that are generated ([Son, Brumley, & Stocker, 2015](#)). With the development of such systems, it is now possible to describe the mechanics of microbial

Table 2 Image Computational Analysis Tools for Microfluidic Microscopy

Programme	Source Code	How to Apply and Work	Last Updated Year	Reference or Contact	Website
Cell tracker	Matlab	Cell tracking and detection Statistical analysis of cell movement	2016	Piccinini, Kiss, and Horvath (2016)	http://www.celltracker.website/
Lineage mapper	Matlab	Overlap-based cell tracking and detection Assign tracks	2016	Chalfoun, Cardone, Dima, Allen, and Halter (2010)	https://isg.nist.gov/deepzoomweb/resources/csmet/pages/cell_tracking/cell_tracking.html
Manual tracking	ImageJ	Cell tracking	2005	Fabrice Cordelieres	https://imagej.nih.gov/ij/plugins/track.html
MicrobeTracker	Matlab	Outlines and segments of cells	2012	Slusarenko, Heinritz, Emonet, and Jacobs-Wagner (2011)	http://microbetracker.org/
MTrackJ	ImageJ	Cell lineage tracking Cell and particle tracking	2015	Meijering, Dzyubachyk, and Smal (2012)	http://www.imagescience.org/meijering/software/mtrackj/
Object tracking software	Matlab	Cell tracking	2003	Nick Darnton	http://www.rowland.harvard.edu/labs/bacteria
ObjectJ	ImageJ	Length and diameter of cell Fluorescent spot detection Spore tracker	2016	Jake Jaff Norbert Vischer Stelian Nastase	https://sils.fnwi.uva.nl/bcb/objectj/

Continued

Table 2 Image Computational Analysis Tools for Microfluidic Microscopy—cont'd

Programme	Source Code	How to Apply and Work	Last Updated Year	Reference or Contact	Website
Oufti	Matlab/ Stand-alone	Cell, spot and object detection	2016	Paintdakhi et al. (2016)	http://oufti.org/
SpotFinder Z	Matlab	Fluorescent signal outline and profile Fluorescent spot detection	2012	Sliusarenko et al. (2011)	http://microbetracker.org/
SpotFinder M Time Lapse Analyser	Matlab	Cell tracking and counting Proliferation and tube assay analysis	2009	Hans Armin Kestler	http://www.informatik.uni-ulm.de/ni/staff/HKestler/tla/
TLM-Tracker	Matlab/ Stand-alone	Outlines and segments of cells Cell lineage analysis Quantification of signal detection	2012	Richard Münch Johannes Klein	http://tlmtracker.tu-bs.de/
TrackMate	ImageJ	Cell tracking Single particle tracking	2016	Jean-Yves Tinevez	http://Imagej.net/TrackMate

locomotion in different environments. These environments can include solid surfaces, flowing fluids and dense cells suspensions. For example, these analytical approaches can be used to define the ‘hydrodynamic signatures’ of swimming bacteria (Elgeti, Winkler, & Gompper, 2015). This signature represents the variation in the fluid flow around the microorganism, as defined by the motility strategy of the organism itself. Dynamic visualisation of motility and the associated hydrodynamic signatures is difficult but can be achieved using a high-speed imaging approach, combined with tracer particles in the medium surrounding the bacteria (Drescher, Dunkel, Cisneros, Ganguly, & Goldstein, 2011). This approach can be used to create cell trajectory visualisations that can indicate the overall swimming path of individual bacteria.

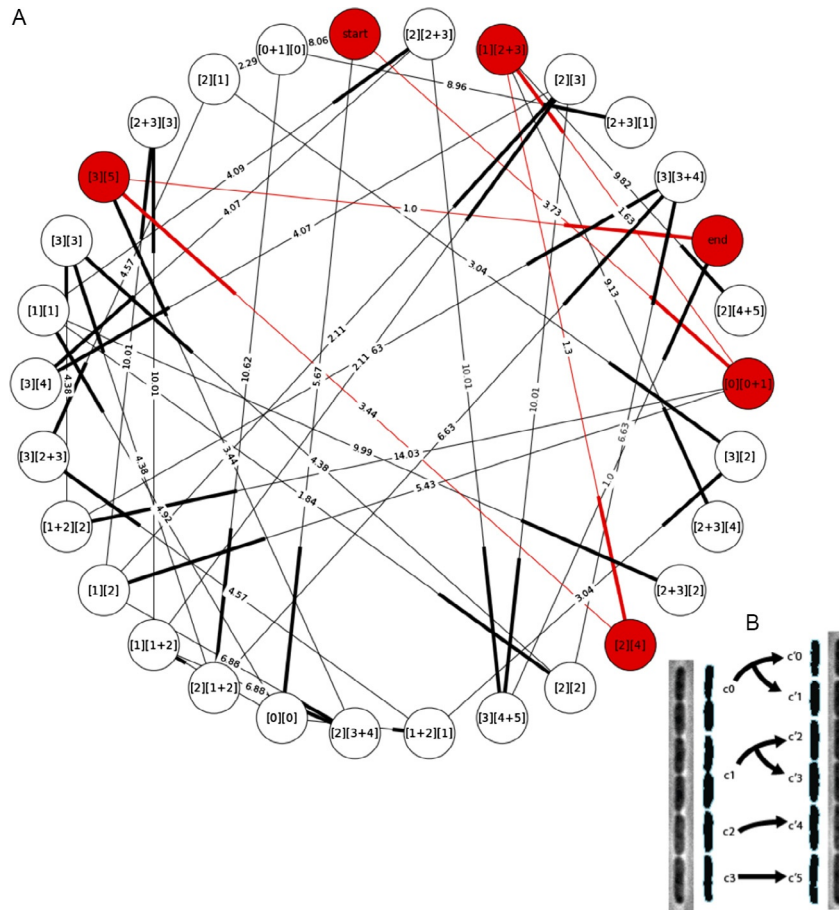
Tracking software has also been used to monitor the growth of bacterial cells grown in channels (such as the Mother Machine; Wang et al., 2010). In the authors’ laboratory, we have developed a system that identifies and monitors single bacterial cells and uses a computational graph-based approach to describe the cell lineages and trajectories. The resulting data allow cell lineages to be analysed computationally using graph traversal algorithms or simply analysed visually (Fig. 4).

In addition to the analysis of cell images, computational approaches play an important part in the control of the environment within the microfluidic chips. Software can be programmed to control the flow of fluid in the system and to control the state of valves which can be used to sort bacterial systems. For example, Li et al. (2014) describe an innovative use of this approach to develop gradients of bacterial growth inhibitors and to automatically measure the response of the bacteria. The system can provide kinetic information on growth inhibition and the dynamics of any morphological changes in the bacteria over a range of chemical concentrations.

Increasingly, droplet-based microfluidics is becoming popular (Zhu & Fang, 2013). In these systems, bacteria are contained in liquid droplets that are suspended in oil. A variety of approaches are then used to characterise the droplets and their bacterial population. These include bright-field and fluorescence microscopy, electrochemistry, laser-induced fluorescence and Raman spectroscopy. One of the advantages of the droplet-based systems is that the computational analysis of the droplets can be used to make decisions about the manipulation of the droplet itself, thus combining analysis and control. The most widely used example of this dual functionality is in cell sorting where droplets containing bacterial cells expressing a particular phenotype are sorted in real time from the rest by physically manipulating the droplet itself (Huang et al., 2015).

6 MICROFLUIDICS FOR THE ENGINEERING OF BACTERIAL SYSTEMS

Microfluidics systems offer an attractive method for automating the process of engineering bacteria and also the process of analysing the resulting bacteria. Microfluidic devices have been described that automate the processes of bacterial DNA

**FIG. 4**

Graph-based analysis of the growth of a single bacterial cell in a microfluidic channels. (A) Graph-based visualisations allow the lineages of cells to be identified; individual cells are nodes in the graph (*circles*) and the edge links (*lines*) indicate the division event leading to the daughter cells for each division. The starting mother cell is indicated at the top of the graph. Therefore, the mother and progeny cells can be defined by following the edges in the graph, either manual or through a software-based analysis. (B) A magnified view of part of the bacterial cell division events in $1\ \mu\text{m}$ channels, that are used to generate the graph representation in (A). The lineage of the cells can be seen as the cells divide to produce daughter cells.

extraction (Matos et al., 2013), RNA extraction (Vulto et al., 2010) and the preparation of DNA libraries (Yehezkel et al., 2016). Droplet technology is proving especially useful in this respect and devices have been proposed to automate many of the steps in the field of synthetic biology (Shih et al., 2015).

The analysis of the phenotype and behaviour of recombinant bacterial systems can also be automated and carried out at a single-cell level (Guo, Rotem, Heyman, & Weitz, 2012). For example, microdroplet-based systems can be used in screening and sorting-based approaches to enrich for bacteria with a given phenotype from a mixed population. In some cases, bacteria have been encapsulated in agarose beads suspended in buffers to facilitate their sorting and manipulation. Eun, Utada, Copeland, Takeuchi, and Weibel (2011) utilised this approach to enrich for rifampicin-resistant mutants from a background nonresistant population. Microfluidic approaches also offer advantages for the study of quorum sensing in bacteria, both natural and synthetic (Fig. 5). The ability to monitor expression dynamics and cellular heterogeneity and stochasticity is also attractive. Ramalho and coworkers describe a microfluidics-based approach for the single-cell analysis of bacterial sender and receiver systems based on *Escherichia coli* engineered with quorum sensing systems derived from *Aliivibrio fischeri* (Ramalho et al., 2016). This approach allowed the dynamics of the gene expression response of receiver bacteria to varying amounts of the quorum sensing inducer N-3-oxo-C6-homoserine lactone (AHL) to

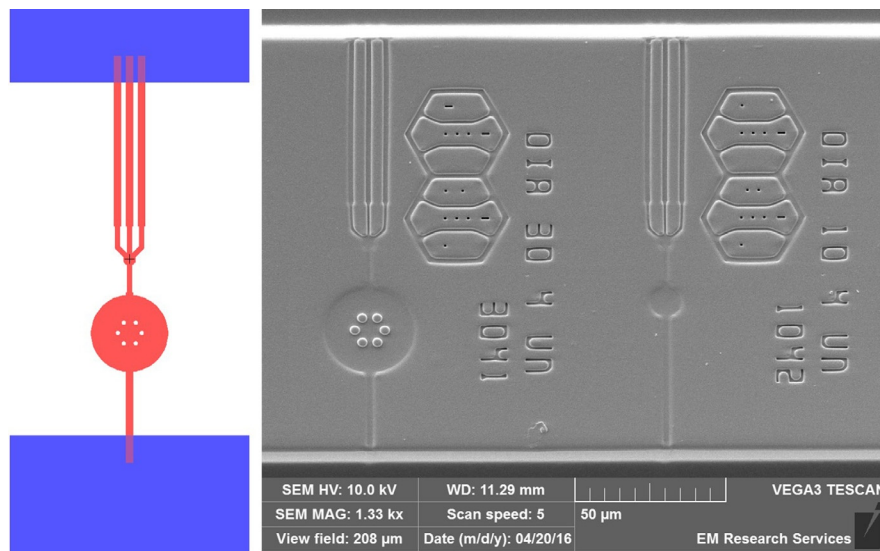


FIG. 5

Left, quorum sensing microfluidics design, produced at Newcastle University. Channels (the circular chamber and the three parallel channels) are designed to trap cells that act as senders and receivers of quorum sensing molecules. Quorum sensing molecules diffuse through the narrow diffusion channels between the cell chambers. The hexagonal structure contains notations about the location of the region of interest as the design is repeated hundreds of times on the same device. *Right*, an SEM image of the design after it has been fabricated using lithography. The device has been backfilled with an oxide material to achieve the final feature size, and then replica moulded from the wafer with PDMS.

be studied. [Danino, Mondragón-Palomino, Tsimring, & Hasty \(2010\)](#) demonstrated the use of microfluidic systems for investigating the collective synchronisation and spatiotemporal waves induced in bacterial populations by quorum sensing molecules ([Danino et al., 2010](#)). They were also able to engineer the genetic circuits to generate synchronised oscillations in growing populations of cells.

7 BIOFILM, MICROBIAL ECOLOGY AND SPECIES–SPECIES INTERACTIONS

Microfluidics has been widely used for studying single-cell dynamics in pure cultures. However, in order to understand how organisms function and respond to stresses and changing environmental conditions, systems for studying mixed communities need to be devised. In a review on current technologies for the analysis of bacterial microenvironments, [Wessel et al. \(2013\)](#) highlight the need for microbiologists to embrace new technologies and realise the limitations of pure culture techniques.

When researching complex bacterial interactions, such as the formation of multispecies biofilms, understanding spatial assembly rules governing how bacteria grow, survive and evolve is important. And an understanding of the relationships between host, biofilm and pathogen is vital for medical and engineering applications. Bacterial communities in nature, industry or medicine are seldom composed of bacteria of a single strain or species, and therefore, understanding the interplay between bacteria of different species is important. In a recent review, [Burmølle, Ren, Bjarnsholt, and Sørensen \(2014\)](#) posed the question ‘Interactions in multispecies biofilms: do they actually matter?’ After reviewing the literature in this area, the authors concluded that the interactions help to determine the function and development of biofilms and, as such, need to be taken into account for medical and industrial biofilm applications. They also highlighted a key challenge for the study of bacterial biofilms, and indeed all microbial systems, namely ‘when diversity increases, so does complexity’. Laboratory methods traditionally used for pure culture-based research, including microfluidics, are now being adapted for assessing such interactions and complexity.

Microfluidics is particularly useful for studying bacterial interactions for many reasons. In microfluidic chips, small numbers of cells can be grown under very specific conditions, with tightly controlled and predictable fluid flow rates and nutrient conditions, including the ability to setup chemical gradients across the chip. Microfluidics also reduces the need for large quantities of growth media or other reagents, due to the small scale of the channels. This is particularly important where specific environmental fluids are used in place of traditional laboratory-produced growth medium, such as the use of saliva to assess the growth and effect of antimicrobials on dental plaque biofilms ([Nance et al., 2013](#)). The use of environmentally relevant fluids as sources of nutrients, and of bacterial cells to seed the community in such experiments, has the potential to allow growth in the laboratory of organisms that

are unculturable in conventional laboratory media. Microfluidic chambers can also be designed with the specific requirements of multiple cell types in mind. For example, they have been used successfully for pathogen–host interaction studies, where the pathogens and their hosts are cultured in separate growth chambers and then exposed to each other in a controlled manner (Kim et al., 2010). The ability of microfluidic systems to be coupled with a variety of microscopy methods means that tracking of cell–cell interactions and biofilm formation in both time and space is achievable. The design of microfluidic systems and their small volumes has the added benefit of allowing for high throughput by simultaneously running multiple samples and controls.

7.1 MULTISPECIES BIOFILM MICROFLUIDICS

In this section, we focus on the latest microfluidic methods for growing and monitoring multispecies biofilms and how microfluidics is being used to determine spatial assembly rules, in both synthetic and natural bacterial communities.

Biofilm research is of great interest to industry. In many industries the build-up of biofilms is problematic, leading to increased costs and adverse health effects. Within the human body biofilms can cause tooth decay and gum disease (Marsh, 1994, 2005), and biofilms on medical devices, such as artificial joints and valves and catheters, can promote infection and affect patient recovery (Song et al., 2013). On the larger scale, biofilms in pipes, for example, those delivering drinking water, can harbour pathogens and potentially cause blockages. In the shipping industry, biofilms on the surfaces of hulls can increase drag and have a negative effect on efficiency, increasing fuel consumption and costs. On the other hand, in some industries, biofilms are required to fulfil a useful role. For example, in wastewater treatment trickling filters, the biofilm on a solid media (such as rock) reduces organic load and is responsible for treating the sewage. Similarly, the biofilms that form on the top layer of a sand filter are critical to the system's ability to purify drinking water.

Microfluidic systems have been used in many studies of biofilms, with the majority of these focusing on pure culture biofilm formation. Although these studies can be informative, they are clearly an oversimplification of what happens in the 'environment' outside the laboratory. There are a number of excellent reviews on the use of microfluidics systems for studying pure culture biofilms (Rusconi, Garren, & Stocker, 2014; Wessel et al., 2013).

A key decision already mentioned earlier is whether to use a commercially available microfluidic device or whether to design, manufacture and run your own system. Commercially available devices have been used to great effect for the study of multispecies biofilms. Indeed, the majority of multispecies biofilm microfluidics studies use these ready-made, all-in-one systems that integrate temperature control, microfluidics chambers and flow control into one easy-to-use system.

By far the most commonly used system for the study of biofilms is the BioFlux system. This system is commercially produced by Fluxion Biosciences and combines

a fluid flow system with a 96-well plate style setup. The cartridges are available in a number of designs (typically a 48-well format), allowing for the simultaneous running of 24 samples or experimental conditions. Each of the two wells in a pair is joined by a microfluidics channel. Channels vary in size depending on the model, but typically have a depth of 70 μm . The wells are used as inlets and outlets for cells and media, as shown in Fig. 6.

Nance et al. (2013) use the BioFlux system along with confocal laser scanning microscopy to study the effects of antimicrobials on dental plaque biofilms. This novel approach, using bacteria from pooled saliva samples to seed the system and cell-free saliva as a growth media led to the growth of biofilms, more closely

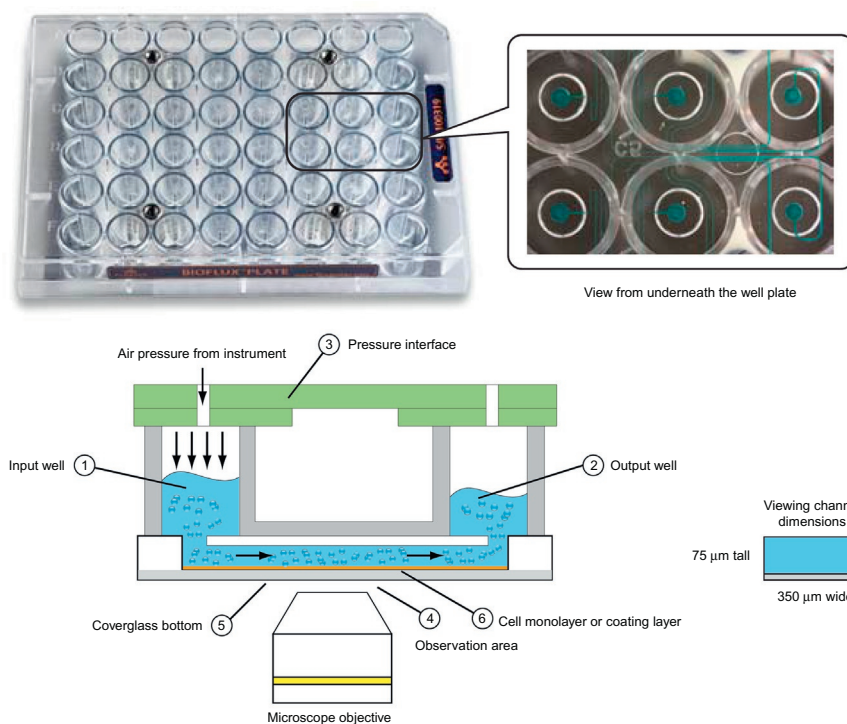


FIG. 6

BioFlux microfluidic device set up. *Top left*, plate format. *Top right*, detail of microfluidic channel between the wells, viewed from below. *Bottom*, schematic of the system. Cells and reagents are loaded into the input well (1), which is connected to the outlet well (2). The BioFlux pressure interfaced applies controlled pneumatic pressure across the top of the wells (3), to drive the flow rate. The sample (6) is observed (4), through the base of the plate (5), a 180- μm glass plate.

Reproduced with permission from Fluxion Biosciences (<http://support.fluxionbio.com/hc/en-us/articles/203648088-BioFlux-Product-Family-Brochure>).

resembling biofilm formation as it is thought to occur on teeth. These biofilms were then challenged with antimicrobial compounds at different concentrations and in situ live/dead staining was used to assess their efficacy. Biofilms were later flushed from the chip, using a high shear stress pulsing flow regime to dislodge the cells from the glass surface. The resulting biofilm was then the subject of bacterial tag-encoded FLX amplicon pyrosequencing (bTEFAP), to confirm that the biofilm that was formed resembles a natural tooth surface biofilm.

Following on from this work, [Kolderman et al. \(2015\)](#) also used the BioFlux system to grow oral biofilms from bacterial cells within pooled saliva samples. In this study the effect of the concentration of L-arginine HCl (LAHCl) on biofilm structure was assessed. This compound has been shown to destabilise biofilm aggregation at high concentrations and mediate cell–cell signalling at lower concentrations. The combination of LAHCl and a common antimicrobial compound found in oral health products, cetylpyridinium chloride (CPC), was tested for efficacy. CPC is known to be an effective antimicrobial, but has difficulty penetrating biofilm surfaces. The study therefore aimed to assess whether using the two treatments in combination would increase biofilm cell death and disaggregation. The ability to image microfluidic channels by confocal scanning laser microscopy means that a wealth of information could be assembled and analysed through image processing. In this study the authors calculated biovolume, average biofilm thickness, biofilm roughness and cell viability (based on live–dead stain fluorescence) using the COMSTAT and IMAGEJ software.

Another variation of the BioFlux system, the ‘Invasion’ plate, was used by [Oliveira et al. \(2015\)](#) to investigate the effects of low concentrations of antibiotics when strains of bacteria are combined. The Invasion plate has two inlets and one outlet per microfluidic channel. This allows media containing antibiotics to be added, with the other channel containing media only, resulting in the formation of an antibiotic concentration gradient. A range of concentrations could be tested in a single experiment and this, coupled with confocal laser scanning microscopy and image processing, allowed the cells’ responses to be recorded directly and analysed. Attachment experiments using the microfluidics device suggested that the cells have an increased likelihood of attaching when antibiotics are present, indicating that biofilm formation was actually promoted by the presence of antibiotics. The authors highlight, however, that the ‘direct assessments of cell division in three-dimensional biofilms is challenging’ and was not done in the study. This is an important limitation of the current microfluidic methods available for analysing biofilm formation. It is challenging to form ‘true’ environmentally relevant biofilms in a microfluidics channel and to design a channel that is also optimal for imaging. Imaging is much simpler when the cells are effectively growing in a monolayer, as in the Mother Machine and the agarose pad microfluidics discussed in [Section 8](#). Once complex three-dimensional structures begin to form, monitoring becomes much more difficult. Potentially, this could be resolved by advances in image processing techniques, a key area for the future development of the field.

7.2 SPATIAL ARRANGEMENTS AND INTERACTIONS

In addition to the study of multispecies biofilms, microfluidics devices have begun to be used to investigate the rules and interactions between bacteria and how this may affect spatial arrangements of bacteria in the environment and in industrial settings. PDMS microfluidics devices are well suited to studying spatial arrangements and interactions, as micron-sized chambers can be designed with varying degrees of connectivity and to hold cells in a range of formations.

By varying the distances between chambers containing three different species of bacteria, [Kim et al. \(2008\)](#) were able to build a stable synthetic community of wild-type soil bacteria with syntrophic interactions. This would not be possible in flask-scale studies, where synthetic communities tend to be unstable. The three bacterial species were specifically chosen because they do not compete for the same nutrients, but each produces compounds critical to the metabolism or survival of the others. The chambers were designed to allow the free flow of metabolites and other chemical cues into the main channel, while keeping the three species at a defined distance from one another. This kind of chip could therefore be useful for gaining a better understanding of communities of organisms, for example, where the biodegradation of pollutants is carried out in multiple stages by different species.

8 CELL CYCLE ANALYSIS AND SIZE HOMEOSTASIS STUDIES

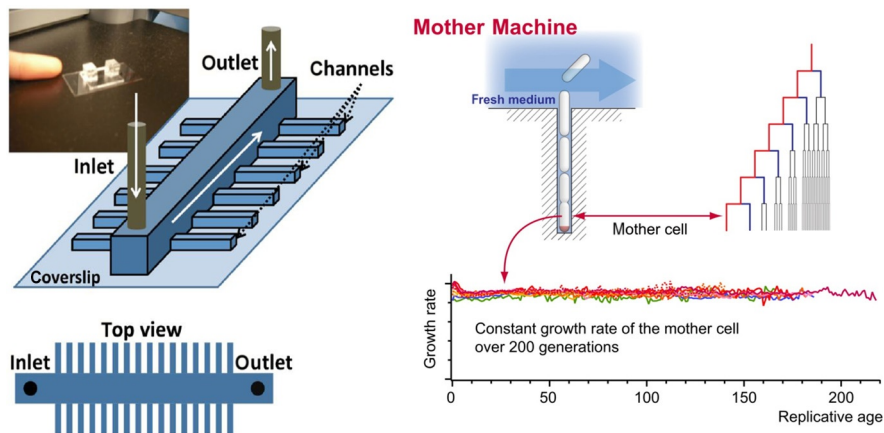
Bacteria propagate by repeated cycles of growth and division, referred to as the cell cycle. During the progression of the bacterial cell cycle, from a new born cell to its division to become two new born daughter cells, three major events occur: DNA replication, chromosome segregation and cell division. Thus, while a new born cell grows in size, it duplicates its genome and segregates the sister chromosomes to appropriate locations in the cell, before making a division septum at the appropriate time and location to generate the two daughter cells. These events occur independently but are well coordinated to ensure the fitness of the progeny ([Adams, Wu, & Errington, 2014](#); [Wu & Errington, 2011](#)).

Cell size control is a fundamental problem for all living organisms. Bacterial cells normally divide only when they are near double the new born sizes; therefore, their size distributions are relatively narrow. It is still not clear how the cell cycle is adjusted so that the cell division is coupled to growth under different growth conditions. For over 40 years, the ‘sizer’ model and the ‘timer’ model were accepted as the possible mechanisms for cell size control. The ‘sizer’ model proposed that bacteria cell size homeostasis was maintained by sensing cell size and that cell division was triggered when a critical size was reached ([Donachie, 1968](#)). The ‘timer’ model, on the other hand, suggested that the timing between DNA replication and cell division remained constant across different growth rates ([Cooper & Helmstetter, 1968](#)). Most of the experimental evidence for these models came from population-averaged data.

Studying size homeostasis requires precise measurement of cell size (length) of a large number of cells grown under steady-state growth conditions. This was achieved recently using microfluidic microscopy. Campos et al. (2014) and Taheri-Araghi, Bradde, et al. (2015) were able to maintain constant growth conditions over a long period of time, so that cells could be imaged and analysed over several generations. The techniques also allowed the authors to monitor and track many cells simultaneously, and to measure cell length from birth to division with high precision at the single-cell level. Both studies show that size homeostasis is maintained through an ‘adder’ rule, where cells sense how much they have grown, and divide when they have added a constant size (volume) since birth, irrespective of the birth size (Campos et al., 2014; Taheri-Araghi, Bradde, et al., 2015). The ‘adder’ rule has now been shown to apply to several evolutionarily divergent bacterial species such as *Caulobacter crescentus*, *E. coli*, *Bacillus subtilis* and *Pseudomonas aeruginosa* (Sauls, Li, & Jun, 2016). Taheri-Araghi, Bradde, et al. (2015) noted that a similar model, termed an ‘incremental model’, was proposed by Koppes and colleagues in the 1990s but was not well accepted. This was due in part to the lack of a ‘comprehensive set of data containing distributions of both size and time between successive cellular divisions’ (Taheri-Araghi, Bradde, et al., 2015; Voorn & Koppes, 1998; Voorn, Koppes, & Grover, 1993), which is exactly what microfluidic microscopy is capable of producing.

Although the microfluidic devices used by the two groups had different designs, both trapped cells in spaces of single layer cell height to ensure monolayer growth. The setups were able to support continuous medium flow, efficiently delivering fresh medium to the growth channels/chambers, at the same time removing cells that emerged from the growth chambers/channels (Campos et al., 2014; Taheri-Araghi, Bradde, et al., 2015; Ullman et al., 2013; Wang et al., 2010). The microfluidic continuous culture device used by Taheri-Araghi, Bradde, et al. (2015), also called the Mother Machine, was shown to deliver culture medium into the channels (by diffusion) much faster (~ 1 s) than the timescale of nutrient uptake (~ 2 – 3 min) (Wang et al., 2010).

The Mother Machine was developed by Jun and colleagues, and the technique has been made accessible through publications in various journals and websites from the Jun lab (Wang et al., 2010; Taheri-Araghi, Bradde, et al., 2015; <http://jun.ucsd.edu/>) (Method 1 in Section 11). The ‘Machine’ is made from PDMS and mounted on a glass slide or coverslip. Instead of having growth chambers like that used by Campos et al. (2014) and Ullman et al. (2013), the Mother Machine features individual growth channels similar to those used by Balaban et al. (2004). The microscale features (i.e. channels and trenches) are replicated onto PDMS from a silicon/SU8 photoresist master mould that contains the negative of the features. Each chip can contain over 1000 growth channels, each approximately $25\text{ }\mu\text{m}$ (L) $\times$ $1.5\text{ }\mu\text{m}$ (W) $\times$ $1.4\text{ }\mu\text{m}$ (D) though the size of the channels can be modified. One end of the channel is closed and the other end opens into a main trench where growth medium flows through (Fig. 7). The cell at the closed end of the growth channel, distal to the trench, is referred to as the ‘mother cell’. The pole of the mother cell

**FIG. 7**

The Mother Machine. The Mother Machine is made with PDMS from the master mould (*left*). The microfeatures on the master mould were made in two steps, the first layer generates the growth channels and the second layer forms the trench. The growth channels are connected to a main trench of varying lengths and widths. Growth medium flows into the device from the inlet, delivering fresh nutrients into the growth channels and ensuring steady-state conditions for all cells (*right*). Cells emerging from the channels are removed by the medium flow into the main trench. The machine allows thousands of cells to be imaged for hundreds of generations.

From fig. 1 and Supplemental Figure—Design of the Mother Machine of Wang, P., Robert, L., Pelletier, J., Dang, W. L., Taddei, F., Wright, A., & Jun, S. (2010). Robust growth of *Escherichia coli*. *Current Biology*, 20, 1099–1103.

that abuts the closed end of the channel is inherited from one generation to the next. The size of the growth channels is designed to match the average diameter of the bacteria to prevent the mother cell from moving around, making image analysis and cell tracking much easier. The Mother Machine is relatively easy to make and use, however, it requires a custom-made master mould (<http://jun.ucsd.edu/>) to produce the PDMS chips, and different moulds are likely to be required for different bacteria or, even different growth conditions under which cell width is expected to change significantly.

To maximise the potential of such devices, several groups have developed microfluidic systems using chips made of agarose instead of PDMS (Ducret et al., 2009; Moffitt et al., 2012; Wong et al., 2010). Various bacteria have been shown to achieve normal growth on agarose (Bergmiller, Pena-Miller, Boehm, & Ackermann, 2011; de Jong, Beilharz, Kuipers, & Veening, 2011; Joyce, Robertson, & Williams, 2011; Lindner, Madden, Demarez, Stewart, & Taddei, 2008; Stewart, Madden, Paul, & Taddei, 2005; Young et al., 2012) and, as a result, agarose has been widely used as a solid support for bacterial growth during microscopy. One of the advantages of agarose over PDMS is its softness, which means that it is less likely to exert

mechanical stress on cells (Mannik, Driessen, Galajda, Keymer, & Dekker, 2009; Moffitt et al., 2012). Agarose is also more permeable to aqueous solutions than PDMS (Moffitt et al., 2012; Qin, Xia, & Whitesides, 2010), an important consideration for experiments that require rapid medium diffusion and small molecule exchange. It has also been reported that uncured PDMS can leach and accumulate in cells, though the effects have not been fully investigated (Moffitt et al., 2012; Regehr et al., 2009).

The agarose-based device designed by the Cluzel lab, called ‘the single-cell chemostat’, has growth channels printed on agarose rather than on PDMS, and the channels open at both ends into gutters or trenches (Moffitt et al., 2012; <http://labs.mcb.harvard.edu/Cluzel/>; Fig. 8). The authors (Moffitt et al., 2012) showed that the

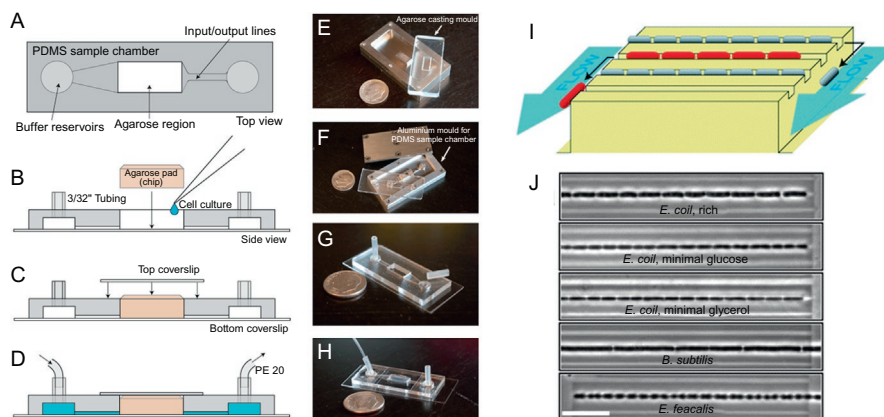


FIG. 8

The single-cell chemostat of Moffitt et al. (2012). (A) Top view of the PDMS sample chamber. (B) Side view of a partially assembled chamber. The bottom cover glass and the Tygon tubing have been added to create a chamber ready for cells and the printed agarose pad. The chamber is connected to the reservoirs at the bottom. Cells are added to the chamber and confined between a patterned agarose pad and a glass coverslip (the bottom coverslip). (C) Once the cells and the pad are loaded, an additional cover glass is laid over the agarose pad, and compresses and seals the device. (D) PE tubing is then used to introduce and remove buffer. (E) A PDMS mould for casting featured agarose pads, and aluminium mould used to fabricate it. (F) A PDMS sample chamber with the aluminium mould used to create it. (G) A partially assembled chamber with Tygon tubing. The PDMS sample chamber is now bonded to a bottle coverslip. (H) A fully assembled device with top coverslip and the agarose chip. (I) Medium flow pass both ends of the channels and removes cells as they emerge from the channels. (J) Soft agarose channels accommodate a range of bacterial morphologies.

Reproduced from Moffitt, J. R., Lee, J. B., & Cluzel, P. (2012). The single-cell chemostat: An agarose-based, microfluidic device for high-throughput, single-cell studies of bacteria and bacterial communities. *Lab on a Chip*, 12(8), 1487–1494 (figs. 3 and S3) with permission from The Royal Society of Chemistry.

agarose channels of this device were able to accommodate a range of bacterial morphologies and, to a degree, different sizes, without perturbing cell morphology or growth. Furthermore, by adjusting the flow rate of the growth medium, this device was also suitable for studying proximity-dependent growth in mixed microbial communities. To generate features on agarose surfaces, the authors fabricated a silicon master containing the positive of the features, which were replicated onto a PDMS intermediate mould with the features in negative. The features were then transferred onto an agarose pad from the PDMS intermediate. This potentially reduces the use of an expensive silicon master and prolongs its life. In addition to a master mould bearing the microfeatures, to make ‘single-cell chemostats’ two further matching custom-made aluminium moulds are required: one for making the PDMS sample chamber that will house the agarose pad and also form medium/buffer reservoirs; the other for making the PDMS mould used for casting patterned agarose pads (chips). These two matching moulds were designed such that the cavity in the PDMS sample chamber is similar in size to the agarose pads but slightly undersized ($\sim 5\%$) in both width and height. The compression needed to fit the agarose pad into the undersized chamber insures that cells are held firmly in place on the bottom cover glass, and that buffer must flow through gutters and not around the pad (Method 2 in [Section 11](#); [Moffitt et al., 2012](#)).

9 CELL SHAPE AND GEOMETRY STUDY

Bacteria come in a variety of different shapes and sizes. However, each genus has its characteristic shape. Interestingly, although some bacterial species do go through morphological changes, this often occurs as a response to environmental cues. It is therefore believed that the shape of a bacterium has biological relevance ([Young, 2006, 2007](#)). For bacteria that have a cell wall, their shapes are defined by the cell wall, which is composed mainly of peptidoglycans ([Cabeen & Jacobs-Wagner, 2005](#); [Errington, 2015](#)). Peptidoglycan is a macromolecule with long, rigid glycan strands covalently linked through short and flexible peptide bridges ([Silhavy, Kahne, & Walker, 2010](#); [Typas, Banzhaf, Gross, & Vollmer, 2012](#)). Many genes involved in the synthesis of peptidoglycan and in bacterial shape determination have been identified, and the genetics of bacterial morphology has also been well studied for several model bacteria. However, the molecular mechanisms of bacterial morphogenesis and cell shape maintenance are still unknown. It is understood that the cell shape is the product of a combination of physical and biochemical processes, but due to technical restrictions it is difficult to study how shape-determining proteins exert controls on cell geometry. Similarly, it is also not clear how a cell regulates its shape in response to external mechanical stress. The advances in microfluidic microscopy have made it possible to address such questions.

The bacterial cell wall needs to have high strength and be sufficiently rigid to resist the internal turgor pressure, yet be dynamic to allow cell growth. The model Gram-negative bacterium *E. coli* and Gram-positive *B. subtilis* are both rod-shaped, which is the most common bacterial shape ([Errington, 2015](#)). While some bacteria

grow by inserting new cell wall material only at the tips or at the division site, *E. coli* and *B. subtilis*, like many other bacteria, grow mainly in length, by synthesising new cell wall at multiple sites in the lateral wall of the cell ('dispersed' elongation) (Daniel & Errington, 2003; Silhavy et al., 2010; Typas et al., 2012). Back in 2005, the Whitesides lab fabricated shaped agarose microchambers and used them to generate filamentous (nondividing) *E. coli* cells of various shapes: crescent, zigzag, sinusoid and spiral (Takeuchi, DiLuzio, Weibel, & Whitesides, 2005). The device also allowed the cells to be released from the chambers into solution. The authors showed that upon being released from the microchambers and grown in solution, the cells retained their shapes for as long as they were prevented from dividing, but reverted back to rod shape when allowed to divide, demonstrating that bacteria could readily deform in response to external mechanical forces but were able to recover their shapes. More recently, several groups have used microfluidic microscopy to further study the effects of external mechanical stress on cell shape in *E. coli* and *B. subtilis* (Amir, Babaeipour, McIntosh, Nelson, & Jun, 2014; Amir & van Teeffelen, 2014; Caspi, 2014; Mannik et al., 2009; Si, Li, Margolin, & Sun, 2015). Using purpose-designed microfluidics chips that imposed geometric confinement or hydrodynamic forces in microfluidic devices, these studies demonstrated both the elastic (reversible) and plastic (irreversible) nature of the growing bacterial cell wall. Interestingly, Mannik et al. (2009) also found that *E. coli* cells were able to penetrate and grow in spaces much smaller than their diameters and readily adapt to the given shapes, even growing as flat, pancake-like cells, whereas *B. subtilis* cells were not able to grow in spaces that were smaller than the diameter of the cell. The lower morphological plasticity of *B. subtilis* is probably due to the difference in cell wall composition between Gram-negative and positive bacteria. Recently, Wu, van Schie, Keymer, and Dekker (2015) also used shaped microchambers (Fig. 9), in combination with drugs such as Cephalexin and A22 that interfered with cell wall synthesis, to sculpture *E. coli* cells into various shapes, and show that Min proteins,

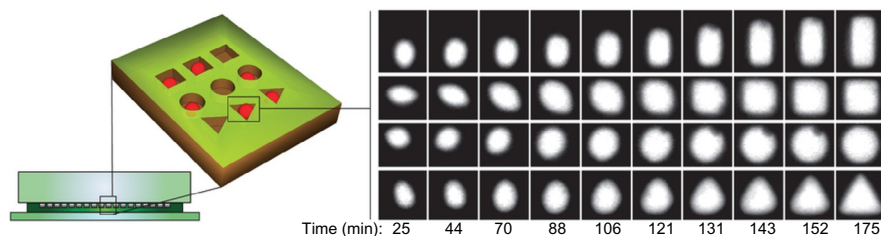


FIG. 9

The 'cell-sculpting' technique of the Dekker lab (Wu et al., 2015). *Left*: schematic of the cell-sculpting device, composed of a microscope coverslip (*bottom*), PDMS microchambers (*middle*, feature side up) and an agarose pad supplemented with nutrient and drugs (*top*). *Right*: cytosolic eqFP670 fluorescence (near-infrared fluorescence with emission maximum at 670 nm) images of *E. coli* cells growing into defined shapes.

From fig. 1 of Wu, F., van Schie, B. G., Keymer, J. E., & Dekker, C. (2015). Symmetry and scale orient Min protein patterns in shaped bacterial sculptures. *Nature Nanotechnology*, 10, 719–726.

involved in determining the site of cell division, could sense and adapt to the morphological features of cells. Like most of the microfluidic chips, these microfeatures were printed from a silicon master mould onto PDMS but while doing so, the authors backed the PDMS film with a coverslip. After loading the bacteria into the microchambers on the PDMS chip, the cells were covered with an agarose pad containing growth medium. The device was then covered with wet tissues and parafilm to prevent drying while allowing air exchange. Although this microfluidic setup did not have continuous medium supply, the device itself is relatively simple to produce and assemble, and would be suitable for experiments that do not require long-term growth.

10 FUTURE PERSPECTIVES

Microfluidic technology has greatly empowered microscopy and allowed us to carry out research that was difficult or impossible to do before. However, with the possibilities come challenges. At present, commercially available microfluidic systems are only suitable for certain types of experiments. From the studies showcased earlier, it is clear that different research needs require chips of different designs. Therefore the application of microfluidics to bacterial imaging is limited by the design and by the availability of microfluidic chips. However, companies are now emerging that provide services for fabricating custom-made silicon moulds and even bench-top fabrication centres. Additionally, new tools are being created for automated design, fabrication and fluid flow control in microfluidics, making microfluidic technology more accessible for laboratories that do not have clean rooms and fabrication facilities.

The recent studies using microfluidic microscopy have provided rare insights into the behaviours of bacteria. However, many fundamental questions remain to be addressed. For example, the molecular mechanism of the ‘adder’ rule that controls bacterial size homeostasis is still unknown; we now know that bacteria adapt their shapes to external forces and constraints but we do not know how the shape change is brought about: is it by altering the distribution or by affecting the function of a protein or proteins? It is also not known how internal mechanical stress influences cell morphology. Microfluidics systems are only just beginning to be explored as methods for the study of multispecies bacterial communities and ecology. This is an area for which rapid prototyping of microfluidic chips to test multiple designs in a short space of time would be of particular benefit.

While the availability of custom-made microfluidic moulds and chips will inevitably widen the application of microfluidics in bacterial imaging, one possible concern is the reproducibility of the experimental conditions by other laboratories. It is therefore important that biologists and micromechanical engineers collaborate closely to create stable and highly definable devices. Similarly, the microenvironments in microfluidic devices can be difficult to characterise, so we also need the expertise of physicists and chemists to help generate devices with optimal medium

flow and molecule exchange. Finally, it is also important that microfluidic devices are designed to be easy-to-use and are applicable to standard microscopy so more research can benefit from the power of microfluidic microscopy.

11 METHODS

As a general consideration, gloves should be used at all time when preparing and handling microfluidic devices.

Method 1 PDMS-based Mother Machine microfluidic device (Adapted from ‘The mother machine handbook’ of Jun Lab, University of California, San Diego, USA; <http://jun.ucsd.edu/>; Taheri-Araghi, Bradde, et al., 2015; Wang et al., 2010)

A. Materials

1. Sylgard 184 Silicone Elastomer kit (containing silicone elastomer base and curing agent).
2. Pentane.
3. Acetone.
4. 50 mg/mL BSA.
5. Growth medium.

B. Equipment

1. Vacuum chamber.
2. Harrick Plasma system.
3. Oven or hot plate for 65°C incubation.
4. Fume hood.
5. Tubing for medium and waste transfer (e.g. 0.023" inner diameter).
6. Needles with Luer-lock interface (22 gauge).
7. Large syringe (e.g. 60 mL) with a Luer-lock tip for the medium.
8. Scalpel.
9. Syringe pump.
10. Magnetic stirrer.
11. Tweezers.
12. Harris Uni-core 0.75 mm punch.
13. Plastic Petri dishes.

C. Preparing supporting PDMS blocks to bolster the PDMS chip.

1. Mix 42 g of PDMS with 4.2 g of curing agent (10:1 ratio) in a standard sized Petri dish. Stir for 30 min and degas using a vacuum chamber until the mixture is clear and bubble-free. During degassing, periodically disturb the vacuum with a brief influx of air to break the bubbles and reduce spillage.
2. Cover the dish and place in a 65°C oven to cure for at least 2 days.

D. Printing micropatterns from the silicon/SU-8 wafer mould onto PDMS

1. Line a large Petri dish with aluminium foil, making sure the foil is flat and smooth. Place the wafer mould in the centre of the dish, feature side up. Blow off any dust then replace the cover to keep clean.
2. For a wafer of 4", mix 9.0 g Silicone elastomer base with 0.9 g curing agent (10:1 to a total of 10 g) for 10 min until the mixture looks milky and bubbly, then degas using a vacuum chamber.
3. Pour the degassed elastomer base/curing agent mixture onto the wafer, tilt the Petri dish to help spread the mixture and cover the entire wafer without spilling.
4. Cover the Petri dish and place at 65°C to cure PDMS overnight.
5. To peel the cured PDMS film off the wafer, first use the tip of a sharp scalpel to separate the two at the edge, then use a pair of tweezers to slowly and carefully peel the PDMS film off the wafer. The peeling motion should be done along the axes of the growth channels. Place the PDMS film in a clean Petri dish.
6. Gently wipe off any uncured PDMS from the back of the wafer with a fibre-free wipe. Do not wipe the feature side. Store the cleaned wafer covered in a clean dish.
7. Cut out the microfluidic chip with a scalpel, leaving about 1 cm along each side of the large channels, and about 5 mm around any fine features. Punch a 2 mm hole at one corner, away from the features, so it can be hung up later.
8. Chemically treat the PDMS chip to remove any residual uncured PDMS. This procedure should be carried out in a fume hood. Wash the PDMS chip firstly in pentane for 2 h with stirring, then transfer quickly to acetone and wash three times, 2 h each wash. 200 mL of pentane/acetone in a 500-mL beaker is sufficient to treat 20 chips. Set the stirring speed to one to five rotations per second and cover the beaker during washing to prevent solvent evaporation. PDMS chips can be picked up with tweezers, and when dropped into the liquid, care should be taken to avoid partially submerging the chip, which could cause deformation.
9. Hang the PDMS chip in fume hood to dry overnight.

E. Assembling the microfluidic device

1. Identify the feature side of the PDMS chip. When looking at the PDMS chip, if the letters and numbers on the chip are readable, then you are looking at the nonfeature side.
2. Place the PDMS chip, feature-side down, in a clean Petri dish. Cut out thick PDMS blocks (from A) and place them in the dish along with the chip, smoothest side up. Then bolster the inlet and outlet areas of the PDMS chip with the PDMS blocks using a plasma cleaner as follows:

First, place the dish into the plasma machine and expose the cleaned surfaces of the PDMS chip and the PDMS blocks to oxygen plasma according to the instruction for your machine (e.g. 15 s at 30 W in a Harrick Plasma system), then immediately lay the blocks onto the PDMS chip (exposed sides facing each other), covering the inlet/outlet areas, and press firmly to expel any air and to ensure complete contact.

Place at 65°C to bond for 10 min. Oxygen plasma makes exposed PDMS and glass reactive, so that covalent bonds form between surfaces brought into contact with one another. The PDMS block should be large enough to cover all fluid inlets and outlets of the chip. Dust and dirt on PDMS surfaces can be removed using scotch tape before plasma cleaning.

3. Locate the trench and punch one hole at each end, to create an inlet and outlet for the fluidic tubing, making sure that the hole goes all the way down through the PDMS chip/PDMS block. Each device can have more than one inlet/outlet. Ensure that the coring tool matches the size of the tubing to be used for supplying culture media (e.g. 0.75 mm coring tool for tubing of 0.023" inner diameter).
 4. Seal the PDMS chip onto a glass bottom dish (e.g. Willco Wells or ibidi μ -Dishes) with no.1.5 glass thickness (170 μ m) or a no.1.5 glass cover slip, again using a plasma cleaner (e.g. exposing the surfaces to oxygen plasma for 15 s followed by bonding at 65°C for 10 min). The glass surface should be cleaned beforehand by wiping with ethanol and then immediately with water to remove residual ethanol. When laying the chip down onto the glass, start at one end then gently tap away bubbles with blunt tweezers.
 5. Bake the plasma bonded PDMS and cover slip or glass bottom dish to set for 10 min.
 6. Inject 10 μ L 50 mg/mL BSA into the device inlet using a standard 10 μ L pipette tip. To avoid introducing air bubbles when injecting the solution, plunge the pipette down to the first stop and hold until the fluid flushes through and emerges from the outlet, then carefully withdraw the pipette tip from the inlet with the pipette still plunged. Seal the inlets and outlets with tape. Incubate the device, uncovered, at 37°C for at least 1 h to passivate the channel walls: the passivation solution will be drawn into the channels via evaporation. The microfluidic device is now ready for loading with the test bacteria. Note that if the device shows any evidence of leakage, it should be discarded.
- F. Inoculating bacteria into the microfluidic device**
1. Concentrate the bacterial culture by centrifugation to, for example, an OD_{600nm} of 4–10. Use a 10 μ L pipette tip to inject 10 μ L of

concentrated culture into the device inlet, again plunging the pipette down to the first stop and holding until the bacterial suspension reaches the growth channels. Carefully remove the pipette tip as before and seal the inlet and outlet with tape. Incubate the loaded device uncovered, at the desired temperature to allow cells to be drawn into the growth channels. Check the device under microscope to monitor loading.

2. Prepare tubing for transferring culture media. Polyethylene tubing (e.g. with a 0.023" inner diameter) can be used. Insert a blunt ended needle into one end of the tubing, then bend the needle to a right angle and twist off the Luer-lock hub. The $\sim 1/2''$ length of metal after the bend will later be inserted into the inlet hole. Insert another blunt ended needle into the tubing and connect it to the medium syringe through the Luer-lock hub.
3. A syringe with a Luer-lock tip can be used to house the growth medium. Install the syringe, containing the medium and connected to the tubing, onto a 'syringe pump'. Pump the medium through the tubing until all the air in the tubing has been displaced (a pump rate of 10–20 mL/h can be used for this step), then remove the tape from the inlet on the PDMS block and insert the bent metal needle end of the tubing into the inlet hole. There should be a drop of medium suspending from the end of the tubing as it is inserted into the inlet hole to prevent any bubble from entering the system.
4. Pump the medium through the device to flush all cells out of the main trench, until the fluid emerging from the outlet appears clear. A flow rate of 5–20 mL/h can be used for this step. Beware that higher rates can generate excess pressure and cause leaks.
5. Turn the pump rate down to the speed appropriate for the experiment (e.g. 1 mL/h). Insert a tube into the outlet hole to collect the waste into a bottle.
6. Mount the device on the microscope stage and begin imaging. Check for leaks continually.

Notes:

1. Add BSA (0.5 mg/mL) to the growth medium.
2. Mother Machine devices need to be optimized for different experiments and different bacteria, e.g., some require larger growth channels.
3. Videos for reference are available at: <http://www.youtube.com/watch?v=RGfb9XU5Oow>; <http://www.youtube.com/user/JunLabHarvard?feature=mhum>

Method 2 Agarose-based microfluidic devices (single-cell chemostats) (Fig. 8; adapted from Moffitt et al., 2012; <http://dx.doi.org/10.1039/C2LC00009A>; <http://labs.mcb.harvard.edu/Cluzel/>)

A. Materials

1. Tridecafluoro-1,1,2,2-tetrahydrooctyl-1-trichlorosilane
2. Copolymer 1: (7–8% vinylmethylsiloxane)–(dimethylsiloxane) copolymer.
3. Copolymer 2: (25–30% methylhydrosiloxane)–dimethylsiloxane copolymer, hydride terminated, 30–50 cst.
4. Modulator: 2,4,6,8-Tetramethyltetravinylcyclotetrasiloxane.
5. Catalyst: Platinum divinyl-tetramethyl-disiloxane; Gelest.
6. Solvent: Hexane.
7. Isopropanol.
8. Acetone.
9. Low melting point agarose.
10. Sylgard 184 Silicone Elastomer kit (containing silicone elastomer base and curing agent).
11. Growth medium.

B. Equipment

1. Vacuum chamber.
2. Harrick Plasma system.
3. Oven or hot plate for 65°C incubation.
4. Fume hood.
5. Spin Coater.
6. Tubing for medium and waste transfer (e.g. 0.023" inner diameter).
7. 30G syringe Luer-lock needle.
8. Syringe with a Luer-lock tip for the medium.
9. Scalpel.
10. Syringe pump.
11. Magnetic stirrer.
12. Tweezers.
13. Biopsy punch (0.75 mm).
14. Glass Petri dishes.
15. Glass coverslips 22 × 50 mm and 18 × 18 mm.
16. Agarose casting mould, made of PDMS from an aluminium mould custom-designed by the Cluzel lab.
17. Aluminium mould for making PDMS sample chambers, custom-designed by the Cluzel lab, and a glass spacer that matches the mould.

C. Casting PDMS sample chamber.

1. Clean the aluminium mould with ethanol and tape around the edge of the mould to form a tray. Use multiple layers of the tape so the 'tray' has four reasonably rigid sides. Make sure that there are no points at which PDMS could leak out. Place in a glass Petri dish.

2. Mix 13.5 g of Sylgard 184 base with 1.5–1.7 g of the curing agent (8:1–9:1 ratio) in a large plastic weighing dish. Degas for 30 min in a vacuum chamber.
 3. Pour about half of the degassed PDMS mixture into the mould. Use a sterile wooden stick to pop any bubble or move them away from the features.
 4. Carefully lower the glass spacer into PDMS, starting at one end and working slowly along to prevent trapping bubbles underneath. Ensure that the spacer is pushed down, resting on the surface of the mould but not touching the round upright features that will form the inlet/outlet ports.
 5. Pour in the remaining PDMS mixture. Place the mould onto the hot plate and leave to cure at 60°C in a fume hood for 24 h.
 6. Remove the tape from the edges of the mould.
 7. To remove the glass spacer, use a scalpel to cut down through PDMS and onto the glass, all the way along both edges of the spacer franked by the round upright features. Cut horizontally through the other two ends of PDMS so that the glass spacer can be released.
 8. Carefully detach the PDMS block from the metal mould, working all the way around to prevent breakage.
 9. Trim the short thick ends and then the long edges until it is slightly smaller than the glass coverslip that the PDMS chamber will be bonded to.
 10. Use a scalpel to remove any excess PDMS from the bottom of the well where the agarose pad will fit. Make sure that the edges are smooth and there are no pieces of PDMS on the surfaces as this will affect bonding and placement of the agarose pad.
 11. Use a biopsy punch to make holes in the centre of the round features for inlet and outlet ports. Push the punch right through and make sure to remove the sliver of PDMS that is pushed out of the hole. Ensure that the coring tool match the size of the tubing to be used.
- D. Preparing the PDMS intermediate, consisting of a thin layer of hard PDMS (h-PDMS) bearing the features, supported by a soft PDMS backing of 3–5 mm.
1. Clean and silanize the master mould wafer by first rinsing the wafer with isopropanol, acetone and isopropanol. Dry with nitrogen and place the wafer on the cap of a small culture tube inside a vacuum chamber in a fume hood. This will allow both sides of the wafer to be silanized, and so prevent PDMS from sticking to the bottom of the wafer. Pipette 20 µL tridecafluoro-1,1,2,2-tetrahydrooctyl-1-trichlorosilane onto a fibre-less wipe

placed in a plastic Petri dish then place the dish in the vacuum chamber. Apply a vacuum for 30 min. Reintroduce air to the chamber very slowly to prevent the wafer from being thrown against the chamber.

2. To make h-PDMS, mix 13.6 g 8% vinylmethylsiloxane–dimethylsiloxane, 72 μ L platinum divinyl-tetramethyl-disiloxane. To the mixture add 0.4 g 2,4,6,8-tetramethyl-tetravinyl-cyclotetrasiloxane, 4 g 25–30% methylhydrosiloxane–dimethylsiloxane and 2 g of hexane. Mix well then degas for 5 min.
 3. Line the spin coater with aluminium foil then mount the wafer on the spin coater. Set the spin coater to spin at 100 rpm for 10 s, and start the machine. Slowly pour the mixture onto the spinning wafer, starting from the centre and move outwards until the wafer is completely covered. Carefully remove the wafer and place on a paper tower in the vacuum chamber. Degas for 3 min. Wipe off any spilt PDMS from the bottom of the wafer as it will interfere with the vacuum seal needed by the spin coater. Return the wafer to the spin coater and spin at 500 rpm for 5 s followed by 1000 rpm for 40 s.
 4. Place the wafer on a wafer holder lined with aluminium foil (dull side up). Bake at 60°C in the oven for 1 h.
 5. To make the soft PDMS (s-PDMS) backing, mix 70 g of Sylgard 184 base with 7 g of the curing agent (10:1 ratio) in a large plastic weigh dish. Degas for 30 min in a vacuum chamber. Gently pour the s-PDMS onto the wafer (on the wafer holder).
 6. Allow the s-PDMS to settle and coat the wafer (~5 min), then place the wafer holder on a 65°C hot plate, cover and leave to cure at 60°C overnight to 24 h.
 7. Carefully flip over the PDMS/wafer/aluminium foil and slowly remove the foil. Run a scalpel along the edge of the wafer to remove any PDMS that has leaked under the wafer.
 8. Slowly and carefully peel the PDMS intermediate from the wafer. Clean the wafer and store covered. Cut out the PDMS intermediate.
 9. Clean the PDMS intermediate before each use by rinsing in double-distilled (dd)H₂O, isopropanol and acetone, followed by isopropanol. Leave to air dry completely.
- E. Printing patterned agarose pad from the PDMS intermediate.
1. Agarose pads are cast in an ‘agarose casting mould’ made of PDMS, onto which the PDMS intermediate is laid (from above) to replicate the microfeatures. ‘Agarose casting moulds’ are made of PDMS and fabricated using a custom-made aluminium mould,

designed by the Cluzel lab to match the mould for making PDMS sample chambers.

2. Melt low melting point agarose (3–5%) in growth medium by heating the mixture at 80°C until completely melted (about 30 min); about 130 μL agarose is needed for each pad. Vortex the agarose/medium mixture frequently to ensure that agarose has melted completely (several rounds of vortexing are needed, returning each time to the 80°C water bath). Leave the mixture for about 15 min after the last round of vortexing to allow bubbles to escape.
 3. Place the cleaned PDMS intermediate on a cold block, feature side up.
 4. In a 30°C warm room, pipette about 65 μL molten agarose onto the PDMS intermediate and another 65 μL into the agarose casting mould. Place the two together and press gently.
 5. Leave to set for 1 h at 30°C.
- F. Assembling the device and loading bacteria.
1. Bond the PDMS sample chamber from C11 onto a large glass coverslip (22 $\times$ 50 mm) using a plasma cleaner (18 W, 20 s, 1000 mTor atmosphere). Bake at 60°C for 10 min to strengthen glass-PDMS bond.
 2. Cut two pieces of Tygon tubing, about 1/4" long and insert them into inlet and outlet. Seal any gaps between the tubing and the surrounding PDMS wall by adding a small drop of PDMS on the inlet and outlet, then baking at 60°C for 30 min.
 3. Plasma clean the PDMS chamber from Step F1, and an 18 $\times$ 18 mm coverslip.
 4. Add 1 μL of bacteria directly onto the glass surface in the PDMS chamber.
 5. Gently separate the PDMS intermediate mould from the agarose pad. Carefully lift out the pad with a clean scalpel and place in the chamber, feature side down. The pad should go in one end first, then gently lower the other end to direct liquid to fill the gutters and push out air bubbles.
 6. If any liquid pools on the top of the gel, remove it gently with a Kim wipe.
 7. Place the plasma-cleaned coverslip over the pad and press gently to seal the PDMS agarose chamber.
 8. Allow plasma bond to set for 15 min (the device can be placed at a temperature appropriate for your bacteria) before introducing growth medium into the device.
 9. Place the PDMS intermediate mould in ddH₂O at 80°C for cleaning.

G. Connecting the device to growth medium

1. Connect the syringe containing the growth medium to tubing and install the syringe on a syringe pump. Load the tubing with medium.
2. Preload buffer reservoirs with medium. This is done by inserting a 30 gauge needle first into the rear buffer reservoir and connecting the tubing to the needle. Run the pump at 30 $\mu\text{L}/\text{min}$ until the reservoir is full. Repeat for the front buffer reservoir.
3. Mount the device on the microscope stage. Run pump at 10–30 $\mu\text{L}/\text{min}$.

Notes:

1. PDMS intermediates can be made with a single layer of s-PDMS if the smallest features (i.e. tracks along which bacteria grow) are greater than 0.5 μm .

ACKNOWLEDGEMENTS

L.E. is funded by an EPSRC Frontiers grant (EP/K039083/1) awarded to A.W., S.L. and L.J.W. are funded by a Wellcome Trust Senior Investigator Award (WT098374AIA) to Jeff Errington and thank him for his support. We are grateful to Suckjoon Jun, Jeffrey Lee and Philippe Cluzel for providing and sharing information on their microfluidic devices, and to Enrique Balleza of the Cluzel Lab for sharing their protocols.

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